



Original article

Mathematical and statistical modelling of electricity demand forecasting using artificial neural networks and SARIMA: Implications for energy supply chain planning

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ABSTRACT

The topic that is under examination deals with the precise measurement of the electrical demand, which is treated as a matter of considerable mathematical and statistical importance. In the context of the developing countries such as Jordan, factors such as the increased sensitiveness to high temperatures, high rates of population growth, and the use of alternative energy resources have made the measurement a matter of considerable importance. The present study introduces an innovative mathematical-statistical model for the accurate calculation of the electrical demand of Jordan during the years 2012–2024. The model particularly highlights the use of artificial neural networks, as well as the use of the seasonal autoregressive integrated moving average model. The study relies on electrical demand data for the years 2012–2024. The model projects the electrical demand for the coming decade. An artificial neural network model is developed for the forecasting of electrical demand, considering various factors, including electrical demand, temperature, humidity, as well as macroeconomic factors, for the Kingdom of Jordan during the years 2012–2024. The precision of the suggested methodology is determined by statistical measurements such as the Mean Absolute Error (MAE), the Root Mean Square Error (RMSE), and the Mean Absolute Percentage Error (MAPE). Simulations were used to create accurate electrical demand forecasts under different heat stress conditions. The results reveal the superior precision of the artificial neural network (ANN) for short-term forecasting when compared to SARIMA, while SARIMA shows robust results for the forecast of electrical demand for the next month. Specifically, the enhancement of forecast precision provides particular advantages to the supply chain, including a 12 % decrease in costs associated with imbalance. The present research aims to present an example of the application of the integration of the artificial neural network methodologies and the statistical models in the energy supply chain of Jordan.

1. Introduction

The issue of electricity demand forecasting was identified as an essential branch of energy economics and SCM due to its direct impact

on the operation, investment, and sustainability of the electric power system. The accuracy of forecasts in developing countries plays an essential role due to various difficulties concerning energy supplies, import dependence, and increased involvement of renewable sources.

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The possibility to determine precise needs for electrical energy surpasses technical characteristics, becoming an essential tool for cost reduction, sustainability, and security [30,53,6]. Traditionally, forecasting of electrical demand is done using linear time series models, namely ARIMA and SARIMA. These models are effective in handling long-term seasonal patterns and trends [15,38]. However, increasing climate change, growing populations, urbanization, and increasing renewable energy have posed difficulties for these models in several parts of the world [42,44]. For example, electrical demand in Jordan is very sensitive to temperature. At the same time, Jordan is targeting increasing renewable energy capacity, particularly solar power [17].

However, there have been limited studies that have focused on electricity demand forecasting in Jordan, considering it from an integrated perspective. Most studies have focused on electricity demand forecasting in the region, considering it in isolation, which means that they have not considered the implications of the precision of the forecasts to the outcomes of the electricity supply chain, such as procurement, reserve, and renewables integration, as discussed in Al-Abri & Okeku [3] and Mystakidis et al. [30]. Most studies have focused on electricity demand forecasting in Jordan, considering it from a particular methodological approach, which means that they have not considered the advantages of linear and non-linear approaches, as discussed in Hu et al. [19] and Ibebuchi [21]. The current research is significant from an application point of view, as it links forecasting accuracy with quantifiable results for an energy supply chain. It therefore links predictive modelling with operational decision-making. It is significant from an academic point of view for researchers, as it adds to the debate on methodology for ANN and SARIMA models, comparing them from an emerging economy point of view, which is often represented in literature [4,47], particularly in data-scarce emerging markets.

The contribution to novel insights is ensured within the scope of the paper due to its dual approach. On the one hand, the paper uses the simultaneous approach for forecasting the electricity demand in Jordan, highlighting the capabilities and limitations of the ANN and SARIMA models for different forecasting horizons. On the other hand, the paper introduces the forecasting approach within the larger scope of Supply Chain Management, thus extending the forecasting approach beyond its traditional statistical boundaries and considering its strategic role in the operations and overall performance of the enterprise [30,7]. This paper uses the Artificial Neural Networks and SARIMA models for forecasting the electricity demand in Jordan for the period between 2012 and 2024 and examines the impact of the climatic, demographic, and economic factors on the demand for electricity in Jordan. The originality of the work does not consist in the proposition of a mathematical hybrid or ensemble model for forecasting, but rather in the innovative approach to the analytical potentialities of Artificial Neural Networks (ANN) and the seasonal autoregressive integrated moving average model (SARIMA), considered from the point of view of an integrated energy supply chain planning model. In contrast to the majority of the existing literature, which focused on the potential for improving the accuracy of the forecast with hybrid ANN-ARIMA and weighted ensemble approaches Zhang et al. [54], the proposed work contributes to the debate on the potentialities for forecasting from the point of view of the connection with the decisions taken in the supply chain for the energy industry. Distinguishing factors for this study include its demonstration of how disparate forecasting paradigms correspond to particular planning horizons, namely, for ANN, managing short-term, non-linear demand volatility, and for SARIMA, managing medium-term, seasonally driven demand. Furthermore, this study goes further than previous research by examining how precision in forecasting affects costs, reserves, and stability, and how these factors relate to integrating renewable energy, which has garnered scant research interest in previous ANN-SARIMA studies, and specifically, in relation to emerging economies like Jordan.

In terms of the specific methods, the research uses time series decomposition, ADF tests for stationarity, and lag selection for SARIMA models, along with the use of ANNs that incorporate weather,

demographic, and calendar factors. Accuracy measures such as MAE, RMSE, and MAPE can be utilized to measure the accuracy of the forecast. Scenario analysis can also provide insight into the supply chain's ability to cope with the increased pressures that could result from temperature changes and the incorporation of renewable energy resources [27]. Essentially, the research links the field of forecasting science and operations research within the supply chain, with the aid of a number of theoretical concepts derived from statistics and theory of machine learning. Accordingly, the research has a number of theoretical underpinnings, including the theory of forecasting, which is a part of the overall methodological framework of the research [35,46]. Supply Chain theory has also been used to examine and create the link between the supply chain and forecasting accuracy [20]. The association between the supply chain and the energy supply has also been used to create the link between forecasting accuracy and the supply chain, an aspect of the topic that needs to be further underexplored. As such, the study of the topic of forecasting and the supply chain is of particular interest to the researcher due to the underexplored nature of the topic within the existing literature.

2. Literature Review

The issue of electricity demand forecasting is considered to be of primary interest to the field of energy economics as well as electric power systems engineering. Traditionally, the issue has been addressed by the application of traditional or linear demand forecasting methods. For this purpose, in this specific paper, autoregressive integrated moving average (ARIMA) models as well as seasonal ARIMA models have been used for the purpose of forecasting the electricity demand. For instance, the seasonal ARIMA models have been found to be quite successful in the context of the Gulf region in the context of electricity demand forecasting [14,37]. [13,16] in their paper published in 2021, used the SARIMA models in the prediction of electricity demand. Besides the application of the SARIMA models in the prediction of electricity demand, other authors have used the ARIMA models in the prediction of electricity demand with remarkable achievements [32]. The main shortcoming of the conventional or linear models in the prediction of electricity demand is their inability to cope with irregular patterns in the demand for electricity, which are affected by irregular changes in climatic patterns and the integration of renewable energy sources [5].

The single most significant factor affecting electricity demand has been identified as temperature. Various studies carried out under differing climatic conditions have confirmed the effect of temperature on electricity demand. Empirical investigations conducted across various regions have proven the association between temperature and the demand for electricity. For instance, investigations conducted across Europe, as highlighted by Boya [9], and other Asian countries, as highlighted by Yao [52], have proven the association between temperature and the demand for electricity. In the Middle East region, extreme temperature conditions have proven to induce exponential growth of the demand for electricity due to the increased demand for cooling, thereby increasing the demand for electricity during the peak demand periods [25]. If we look at other studies, despite the limited breadth of study on the matter, the effect of humidity on electricity demand has been acknowledged. This can be illustrated with the following example: the study [26] on the effect of temperature and humidity on the demand for electricity showed that the inclusion of the effect of humidity produces better model performance compared with the inclusion of the effect of temperature alone.

It has been recognized that the patterns and trajectory of the use of electricity are influenced by certain demographic and economic factors. It has been noted that the patterns and trajectory of the use of electricity are directly influenced by the rate of growth of the human population, both in the residential and commercial sectors. It has also been noted that the patterns and trajectory of the use of electricity are positively influenced by the gross domestic product (GDP). It has also been noted

that the patterns and trajectory of the use of electricity have a markedly positive impact on the economy and the rate of economic growth in developing economies such as India and China [50]. Factors that have been noted to have an impact on the patterns and trajectory of the use of electricity, as noted in the Jordan case study, are the rate of growth of the human population and the extent of economic diversification. The integration of a growing percentage of RE into the grid is forecast to create variability in the NL, which is a challenge to the accuracy of the forecast. Research carried out in Europe and North America highlights the significance of accurate forecasting in the integration of RE, owing to the role played in the balancing supply and demand. Research carried out in Morocco and Egypt highlights the adverse impacts of forecasting errors in RE integration, which result in costly peaking power supply. In the context of Jordan, the accuracy of demand forecasting is essential to ensure the balancing supply and demand, especially in the context of the expansion in the capacity for the supply of solar power post-2015 [31]. The role played by the ARIMA and SARIMA models is essential for the integration of RE, which incorporates the concepts of autocorrelation, differencing, and moving averages for the detection of trends [48,49].

Such methodologies have proven to be adequate for handling datasets that have well-established seasonality and trending patterns, as these methodologies have been extensively used in energy forecasting. The linear nature that these methodologies assume to be present in the datasets constitutes the major limitation that these methodologies face, especially in the event that the structure that the datasets follow exhibits sudden change or if the relationship that exists between the weather conditions and the load exhibits a nonlinear relationship. The application of Artificial Neural Networks (ANNs), on the other hand, has also proven to be applicable to load forecasting as a result of the nonlinear relationship that it can approximate. A study done by Park et al. in 1991, as well as Highley, Hilmes [18], sought to establish the potential that exists for the application of ANNs to be better than the application of time-series analysis techniques to load forecasting. The application of the ANN has also been further explored to establish its potential under different conditions, as discussed in the study done by Buitrago, Asfour [10], as well as Tarmanini et al. [43]. The application of the ANN to the forecasting of load that occurs in the short term has proven to be the most applicable as a result of the effect that the weather conditions have on the load.

To check if the forecast is accurate or not it measured through various parameters such as the mean absolute error, root mean squared error, and the mean absolute percentage error. Scholarly literature pertaining to energy forecasts emphasizes the need to use a variety of parameters to measure the bias, variance, and percentage error of the forecasts [45]. The preliminary work includes performing unit root tests such as the Augmented Dickey-Fuller test and analysing the characteristics of the autocorrelation function as a prerequisite to build an accurate SARIMA model [22]. Research on different forecasting techniques applied to energy systems has successfully applied the methods to ensure the correct lag specification for the model. This methodological approach can help to ensure the accuracy of the forecast, which might even help to avoid instabilities in the forecast. Scenario analysis methods are being applied to assess the accuracy of the forecast [28]. Xiang et al [51]. highlighted the importance of applying the scenario analysis method to grids where the generation is mostly from renewable resources. In addition to this, the method is being applied to the Middle Eastern countries to assess the accuracy of the forecast for heat waves and the integration of renewable resources into the grid. This methodological approach can help to assess the accuracy of the forecast methods applied to the supply chains [36].

2.1. Research gap

The aforementioned literature indicates that, on one hand, ANN and SARIMA possess certain advantages when applied to specific contexts, where ANN is found to perform better under short-term, nonlinear

circumstances, and SARIMA is found to provide strong forecasts under long-term circumstances, especially when dealing with seasonal data. On the other hand, the aforementioned studies highlight that climatic, demographic, and economic factors impact electricity demand, although there are three major gaps that need to be addressed, where the first is that there is a lack of studies conducted on the Jordanian context that address the integration of demand forecasting with energy supply chain management, especially when dealing with the aforementioned areas of procurement, scheduling, and integration of renewable resources, whereas the second gap is that, when conducting electricity demand forecasting, the aforementioned studies rarely address how improvements to forecast accuracy can lead to improvements to operational costs, whereas the third is that, when dealing with emerging economies, there is limited application of scenario-based stress testing to assess the robustness of forecasting tools when dealing with circumstances of climate perturbation and integration of renewable resources.

2.2. Conceptual framework & hypothesis development

The framework for this paper will be based upon an approach that integrates demand forecasting models and the management of the energy supply chain (Fig. 1). The determinants of the demand for electricity in Jordan include environmental, demographic, economic, as well as structural determinants. The determinants are considered as independent variables that influence the demand for electricity in Jordan. These determinants include temperature, humidity, population, the rate of economic growth, as well as the fraction of renewable energy incorporated in the entire power production scheme. The methodology of the paper will involve an analytical technique that makes use of two different models. These two models will utilize the Artificial Neural Network model as well as the ARIMA/SARIMA model, which will function simultaneously to reduce forecast errors. The framework has two levels: the primary level will concentrate on forecast precision, which can be measured through the Mean Absolute Error (MAE), Root Mean Square Error (RMSE), as well as the Mean Absolute Percentage Error (MAPE). The second level will concentrate on the effects of forecast precision on the energy supply chain, where the expectation is to have a positive effect on the supply chain.

The theoretical framework consists of three components. These components are the theory of forecasting, supply chain management, and energy planning. Regarding the theory of forecasting, the research uses ARIMA/SARIMA models and ANN to forecast the electricity demand. In addition, the accuracy of the forecast, which acts as a mediator variable, has a basis in the theory of forecasting and uses MAE, RMSE, and MAPE as measures of accuracy. Contrariwise, the theory of supply chain management provides an explanation of the impact of the accuracy of the forecast models on the operations of the energy industry. In particular, the use of lean and agile strategies within the SCM reinforces the importance of managing the so-called bullwhip effect, described as the “amplification of demand variations within the energy SC.” The energy planning literature, with its emphasis on Integrated Resource Planning and the theory of Economic Dispatch, positions demand forecasting centrally within the determination of procurement strategies and the balance of traditional and alternative forms of energy generation. Indeed, more accurate demand forecasting allows for the minimization of reserve margins, the optimal production of alternative energy modalities, and the minimization of costly interruptions during periods of maximum demand. All in all, these theories collectively provide an outline of the order of events from the determinants of demand forecasts to the outcomes in the supply chain. This framework considers the forecasts of potential demands as an operational strategy in Jordan’s demand-driven, cost-effective, and resilient supply chain. Therefore, the study was informed by the following hypotheses:

H1: Artificial Neural Networks (ANN) provided significantly higher forecasting accuracy compared to traditional ARIMA/SARIMA models in predicting short- and medium-term electricity demand in Jordan.

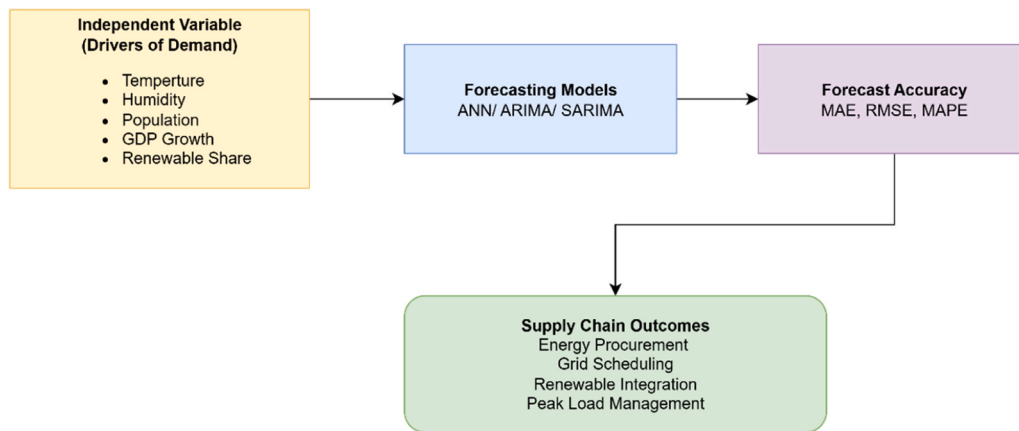


Fig. 1. Conceptual Framework of The Study. Source: Author.

H2: The incorporation of weather variables (temperature and humidity) and macroeconomic indicators (GDP growth and population) significantly improved the accuracy of electricity demand forecasts.

H3: Accurate electricity demand forecasting had a significant impact on optimizing supply chain planning in the Jordanian energy sector, particularly in energy procurement, grid load scheduling, and renewable energy integration.

H4: Forecasting errors were positively associated with peak load disruptions, implying that greater prediction accuracy contributed to operational agility and demand-supply synchronization.

3. Methodology

The study follows an applied, quantitative, analytical, and explanatory approach, which aims at clarifying the relationship between various factors, as exemplified by the electricity demand pattern in Jordan, temperature, population growth rates, economic activity, etc. It is noteworthy, however, that the approach also includes a comparative dimension, as exemplified by the use of Artificial Neural Networks (ANN) as well as the Autoregressive Integrated Moving Average/SARIMA model, which were used in the forecasting of the electricity demand pattern. Significantly, the study also emphasizes the need for evaluating the accuracy of the forecast in relation to the activity of the supply chain. The study depends solely on secondary data collected from reliable sources. To be precise, the data on the patterns of electricity demand at the hourly, daily, and monthly levels are collected from the National Electric Power Company (NEPCO) and the Ministry of Energy and Mineral Resources in Jordan. Besides this, the study will be dependent on the following variables collected from the Department of Statistics (DoS), Jordan: the growth rate of the country's GDP, the level of production in the industry sector of the Jordanian economy, and the population size of the country. Data on weather variables such as temperature, humidity, and wind speed are collected from the Jordan Meteorological Department for the period from 2012 to 2024.

The study population refers to the complete set of records on the electricity demand in Jordan, including all consumer types, ranging from residential to governmental bodies. It was not feasible to study the patterns at the level of individual households or entities, so the study was done at a level that represents the actual patterns observed at the national grid level. The population, in this case, refers to the complete set of records on the electricity demand in Jordan ranging from the year 2012 to the year 2024, alongside the relevant features.

In the context of time series analysis, the sample size is generally influenced by the availability of historical data rather than random sampling. Nevertheless, a statistical basis for the study was achieved using Cochran's formula for continuous data, where the mean and standard deviation of the values corresponding to the monthly demand

formed the basis for the calculation. Based on the 95 % confidence level and an acceptable error rate corresponding to five percent of the average demand, the minimum sample size for the study was established to be forty. In reality, the dataset comprised 156 observations for the monthly data and over 110,000 records for the hourly data over a period of twelve years, which far exceeded the minimum sample size for the study. This ensured the robustness of the statistical basis for the study. In addition, the dataset from January 2012 to December 2024 was divided chronologically to ensure there was no leakage of data for the train, test, and validation phases. In this regard, the dataset from January 2012 to December 2022 was utilized for the training phase, while the dataset from January 2023 to December 2024 was utilized for the test phase for all the forecasting methods. This chronological splitting was applied to all the forecast methods.

The data for electrical demand was obtained from the National Electricity System Operator of Jordan. The electrical demand is given in megawatts. The data is available from January 2012 up to December 2024. The data is available at an hourly level. The data is then aggregated at different time periods depending on the horizon of the forecasts. The variables for weather-related factors such as temperature and humidity were obtained from the National Meteorological Authority of Jordan. The data is available at an hourly level. The values of temperature and humidity are given in degrees Celsius and percentages, respectively. The values of macroeconomic variables such as growth rate of population and GDP growth rate were obtained from government statistics. The values of these variables are available at an annual level. The final variables for analysis are historical values of electrical demand, lag values of electrical demand, temperature, humidity, calendar variables such as day of week and month, growth rate of population, growth rate of GDP, and renewable share. The missing values in the datasets were less than 1.5 %. The missing values were replaced using linear interpolation. The datasets were then normalized using min-max normalization. The missing values in the datasets were less than 1.5 %. The missing values were replaced using linear interpolation. The datasets were then normalized using min-max normalization. The missing values in the datasets were less than 1.5 %. The missing values were replaced using linear interpolation. The datasets were then normalized using min-max normalization.

The descriptive features of the dataset are shown in Table 1. On average, the demand on the grid over the period has been 2850 MW with a standard deviation of 510 MW. The minimum demand has been 1480 MW, while the maximum demand has reached 4620 MW. The average temperature has been 21.9 ° Celsius with a minimum temperature recorded to be 5.3 ° Celsius and a maximum recorded temperature of 39.5 ° Celsius. The population of the country has steadily increased over the years. In 2012, the population stood at 7.8 million. In 2024, the population increased to 10.9 million. On average, the gross domestic

Table 1
Description of Population (Electricity Demand in Jordan, 2012–2024).

Variable	Description	Unit	Mean	Std. Dev.	Min	Max
Electricity Demand	Aggregated load demand	MW	2850	510	1480	4620
Temperature	Average daily temperature	°C	21.9	7.2	5.3	39.5
Population	Mid-year total population	Millions	9.5	0.7	7.8	10.9
GDP Growth	Annual real GDP growth rate	%	2.4	1.2	-1.9	4.3
Renewable Share	Share of renewables in electricity generation	%	22.7	9.4	3.0	36.5

Source: Author

product has grown by 2.4 % with occasional declines due to economic shocks. However, the share of renewable energy has increased remarkably. On average, the share has been 22.7 % with a maximum recorded value of 36.5 % in 2024.

The study also included the concept of both independent and dependent variables. In this respect, the dependent variable and the target of the forecast was electricity demand. On the other hand, the independent variables included factors such as temperatures, humidity, and wind speed to account for the effects of the weather. Other factors included the growth rate of the gross domestic product and the population to account for the effects of the economy and the population. In addition, the study included the rate of renewable sources of energy to account for the changes in the structure of the economy.

The electricity demand was specified in megawatts and was made equivalent to the peak demand. Weather variables are collected as daily averages through the Jordanian Meteorological Department. Similarly, macroeconomic variables are collected through official government statistics. Population figures are based on mid-year values. Similarly, renewable shares are specified as the proportion of electricity produced through renewable energy sources, as reported through the Ministry of Energy. These variables are selected for their temporal comparability and matching with international forecasting research.

The analytical steps began with the preprocessing of the data. The stationarity of the time series was checked using the Augmented Dickey Fuller test. Additionally, the analysis of the Autocorrelation Function and the Partial Autocorrelation Function was carried out for the data. For the ANN model, the data was normalized to ensure the achievement of the best possible results and to avoid the risk of computational bias. For the forecast generation, the ARIMA and the SARIMA models were applied to the dataset to identify the linear trends in the data, while the ANN model, which is a type of feed-forward network known as the Multi Layer Perceptron, was applied to the dataset to identify the nonlinear trends within the dataset. For the ANN model, the dataset was divided into the training set and the testing set in the proportion 80:20 to ensure the reliability of the evaluation carried out for the model. For the accuracy assessment of the developed models, the Mean Absolute Error, Root Mean Square Error, and Mean Absolute Percentage Error metrics were applied to the dataset. Additionally, a sensitivity analysis was carried out to assess the variations in the demand for the product using the scenario simulations. The analysis carried out for the ANN and the SARIMA model was further assessed to identify the implications for supply chain operations. In the context of the study, the word "integrated" refers to the ANN and the SARIMA model being applied in a comparative and decision-making context to arrive at a result for the dataset. In the context of the study, the ANN model is applied for short-term forecasting, while the SARIMA model is applied for the medium-term forecast. The result generated from the integrated model refers to an optional analysis carried out to assess the robustness of the result generated from the ANN and the SARIMA model using a simple form of

forecast averaging.

The Artificial Neural Network model was developed and implemented as a multilayer perceptron model consisting of an input layer, two hidden layers, and an output layer containing a single output neuron for electricity demand. The input layer consisted of lagged demand, temperature, humidity, calendar variables, and macroeconomic variables. The hidden layers consisted of 64 and 32 neurons, respectively. The ReLU activation function was used for these hidden layers. The output layer used a linear activation function. The Adam optimization algorithm was used for training with a batch size of 32 and a learning rate of 0.001. The Mean Squared Error was used as the loss function. Normalization of input variables was done using min-max scaling. Overfitting was prevented using early stopping. The input variables were divided into training, validation, and test sets. The training set consisted of 70 %, and each of validation and test sets consisted of 15 %. For the seasonal autoregressive integrated moving-average model, the parameters (p,d,q)(P,D,Q,s) have been selected following the Box-Jenkins approach. Tests for stationarity have been performed using the Augmented Dickey-Fuller test, which guided the choice for the first-order and seasonal differences. Using the ACF and PACF plots, the order for the autoregressive and moving-average terms has been selected for the SARIMA ((1,1,1)(1,1,1,12) model for monthly demand. Further, the adequacy of the model has been tested for the residuals, and the tests for the presence of any autocorrelation have been performed. Normal tests have been performed for the residuals.

All these analyses have been carried out by utilizing Python as the programming language, specifically utilizing version 3.9. The Artificial Neural Network (ANN), for instance, has been created by utilizing TensorFlow and Keras, whereas the SARIMA model has been estimated by utilizing stats. The Artificial Neural Network model has two hidden layers, comprising 64 and 32 neurons, respectively. The model has utilized 'ReLU' as its activation function. The model has utilized 'Adam' as its optimizer, and its learning rate has been fixed at '0.001'. The batch size has been fixed at '32', and a maximum of '200' epochs have been utilized. The model has utilized 'early stopping' and its patience has been fixed at '10'. To ascertain stability, these models have been trained by utilizing fixed random seeds. The SARIMA model has utilized maximum likelihood procedures for model estimation, and parameters have been chosen by utilizing techniques for minimizing 'AIC'. In addition, 'residual diagnostics' have been carried out for model adequacy. The model has utilized 'out-of-sample' testing for model accuracy, wherein 'MAE', 'RMSE', and 'MAPE' have been utilized for model accuracy. The model has followed all standard procedures for model accuracy, and all these procedures have been followed for model accuracy.

It should be noted that despite the fact that the study was based on secondary and aggregated sources of data, the ethical norms were adhered to with respect to the entire process of the study. All the sources of the data were official and available to the public. There was no breach of the privacy and confidentiality of the sources of the data. In addition, the sources of the data were duly acknowledged to ensure the integrity of the study. There was also the full disclosure of the findings of the study, and this was done with full impartiality and no manipulation of the findings of the study.

4. Results

4.1. Electricity demand patterns in Jordan

The analysis of the pattern followed by the demand for electricity from 2012 to 2024 suggests an upward trend for the demand for electricity. There are seasonal variations in the pattern followed by the demand for electricity during the respective months. Two major peaks are identified for the demand for electricity during the respective months. The major peak for the demand for electricity is observed during the summer months from July to September, which can be related to the increased demand for air conditioners during the respective period. The

second major peak for the demand for electricity is observed during the winter months from January to February. The trend for the demand for electricity from 2012 to 2019 followed an upward pattern at an average rate of 2.2 %, which declined during the year 2020 because of the economic slowdown. From 2019 onwards, the trend for the demand for electricity followed a moderate pattern. From the analysis, the temperature is identified to have the major effect on the demand for electricity, with a correlation greater than 0.70 at the daily level. This confirms the nonlinear dependence between the demand for electricity and the respective climatic conditions.

The pattern followed by electricity demand in Jordan for the period 2012–2024 reveals an increase in demand while also showing strong variations. The daily mean demand for electricity reaches maximum figures during summer. The maximum value was observed on 28 August 2024, i.e., 4712 MW. The Augmented Dickey-Fuller test for testing for stationarity reveals that the original series was found to be non-stationary. The ADF value was -1.88 , while the p-value was observed to be 0.64. After differencing and seasonal differencing at lag 12, stationarity was observed. The ADF value was -5.67 , while the p-value was observed to be very low, i.e., less than 0.001. The ACF and PACF analysis also revealed that an ARIMA model was appropriate.

The time series of the actual demand and the forecasts for the year 2023 are depicted in Fig. 2. In addition, the monthly mean loads and the peak event experienced in the month of August are also indicated. From the graph, the seasonality of the demand and the similarity between the ANN and SARIMA models are evident.

4.2. Model performance and forecast accuracy

The forecasting exercise compared artificial neural networks (ANN) with seasonal autoregressive integrated moving average (SARIMA) models for three forecasting horizons: day-ahead (hourly), week-ahead (daily), and month-ahead (monthly) forecasting. For shorter forecasting horizons, artificial neural networks demonstrated better accuracy than the other models. It was successful in capturing the non-linear relationship between temperature and electrical demand, as well as the impacts of holidays. For longer forecasting horizons, SARIMA models showed better performance compared to artificial neural networks. This is because SARIMA models are effective in handling seasonal and trend effects.

Results revealed that artificial neural networks (ANN) performed considerably better than SARIMA for day-ahead and week-ahead forecasting, as revealed by the Diebold–Mariano statistical tests. However, SARIMA performed marginally better than ANN for month-ahead forecasting. A combination of the models further reduced the day-ahead mean absolute percentage error (MAPE) to 3.82 %.

SARIMA proved to be effective in capturing the linear as well as the seasonal dependencies present in the demand series. The most robust model for the monthly demand series proved to be the SARIMA (1,1,1) (1,1,1) [12] model. While the long-term forecasting capabilities were strong for the SARIMA model, its performance became less stable in the face of short-run demand shocks. The error distribution also proved that the error distribution for the SARIMA model became wider and more dispersed, as shown in Fig. 3. (Table 2)

4.3. Forecasting of peak loads

Moreover, the accurate forecasting of extreme loading events became critical to the grid reliability. The annual peak demand for the year 2024 occurred on the 28th of August, with the actual demand amounting to 4712 MW. The artificial neural network (ANN), while forecasting this extreme loading event, underpredicted the actual demand by merely 151 MW, showing a low error rate of merely 3.2 %. On the other hand, the underprediction by the SARIMA model amounted to a higher error rate of 230 MW, or 4.9 %. Out of the top fifty highest peak load hours observed within the period of the year 2023–2024, the artificial neural network achieved a low mean absolute percentage error rate of merely 2.9 %, showing a marked improvement over the error rate achieved by the SARIMA model, which amounted to a higher error rate of merely 4.1 %. Moreover, the artificial neural network model shown schematically in Fig. 4 shows the structure formed within the network. The feature importance plots indicated that the lagged load (40 %) and temperature (35 %) features were the most important features within the dataset.

Moreover, a comparative analysis of these schemes in a side-by-side format (Fig. 5) reveals the relative strengths of these approaches. The performance of the proposed ANN scheme was found to be better in short-term forecasting, as it attained a peak hour MAPE of 2.9 %, compared to SARIMA's better accuracy in month-ahead forecasting.

4.4. Scenario simulations and sensitivity analysis

Two scenarios have been set up in order to test the robustness of the models for different demand profiles. These scenarios include the heat stress scenario, which consists of a uniform temperature increase of $+2$ °C for the maximum summer temperature for the months June to September, which is consistent with the climate change projections for the Eastern Mediterranean region. This has been implemented for the temperature component while keeping the rest constant. Another scenario includes the increase in the portion of renewable energy within the total energy mix for the region. This consists of an increase in the total portion of solar and wind power within the total electrical energy mix by

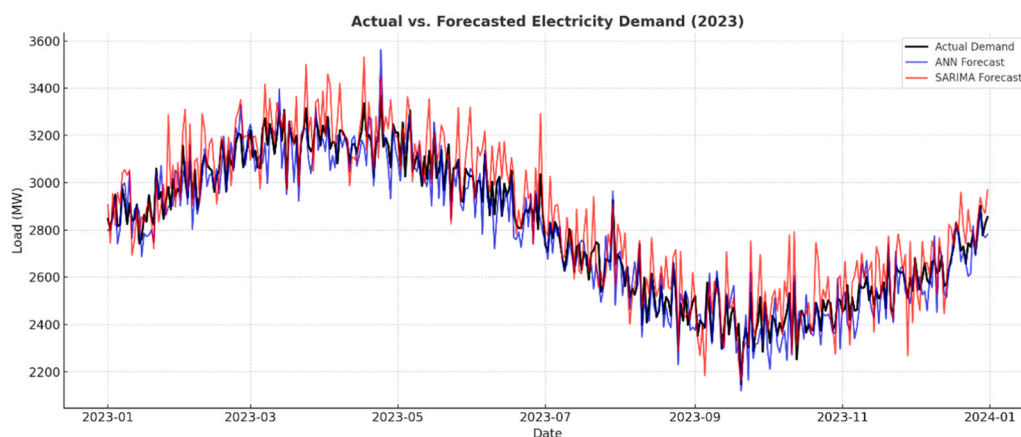


Fig. 2. time-series of actual demand versus forecasts for 2023.

Source: Author

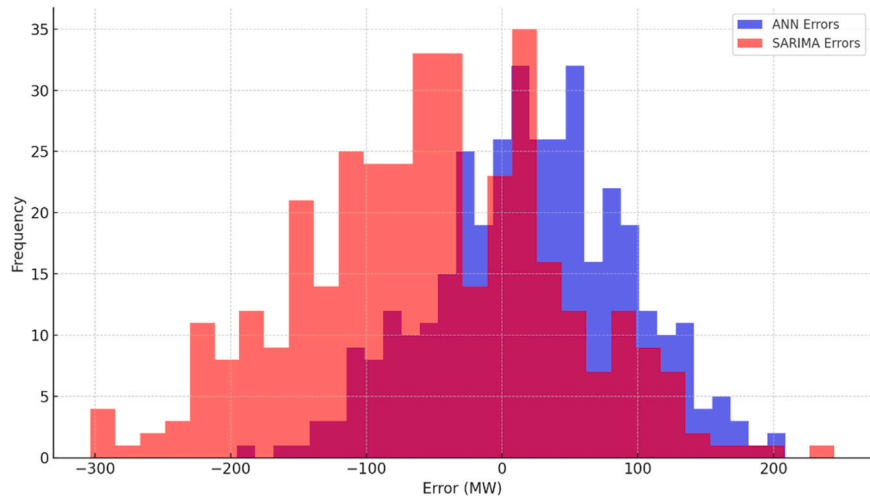


Fig. 3. Source: Author.

Table 2
Main Variables of the Study.

Variable	Type	Measurement Level	Role in Study
Electricity Demand	Dependent	Continuous (MW)	Forecast target
Temperature	Independent	Continuous (°C)	Predictor
Humidity	Independent	Continuous (%)	Predictor
Population	Independent	Continuous	Predictor (macro factor)
GDP Growth	Independent	Continuous (%)	Predictor (macro factor)
Renewable Share	Independent	Continuous (%)	Predictor (supply factor)

Source: Author

10 %. This has been implemented for the models by increasing the net profiles for the day for the scenarios.

Two scenarios were designed for stress testing. The accuracy was maintained for scenario 1, where a + 2 °C increase in summer temperature was applied uniformly. As a result, an increase in overall peak demand was observed by around 7 %. The accuracy level was maintained for ANN, where mean absolute percentage errors increased marginally from 3.9 % to 4.6 %. For SARIMA, errors increased significantly from 4.7 % to 5.8 %. For scenario 2, where solar and wind were

increased by 10 %age points, ANN-based scheduling was observed to have stable forecasts for net load demand. The level of renewable curtailment was reduced by 6.3 %.

The results suggest that electricity demand in Jordan is significantly seasonal and nonlinear with respect to climate. In particular, it was found that ANN outperformed SARIMA for short-term forecasts, especially for peak prediction, whereas SARIMA was slightly superior for long-term planning. As such, the significance of improved forecasts is that they would help minimize costs, reduce stress, and enable the integration of renewables. Overall, it was made clear that forecasting, as a discipline, is not merely limited to statistics; it is, in fact, a strategic approach to building a resilient and responsive energy chain (Table 3).

Hypothesis. Testing Results

The basis of this hypothesis is that ANN will produce considerably higher values of forecasting accuracy compared to ARIMA/SARIMA models. The analysis carried out for the period of 2023–2024 showed that ANN produced an average absolute percentage error (MAPE) of 3.94 % for day-ahead forecasts, which is considerably lower than SARIMA’s 4.67 %. The Diebold-Mariano test for absolute errors of forecasts showed that the difference between ANN and SARIMA is significant at a 5 % level of significance, given that the DM statistic is 2.41 (p-value of 0.016). Similar results were obtained for week-ahead forecasts, where ANN outperformed SARIMA with MAPE values of 3.86 % and 4.09 % for ANN and SARIMA models, respectively. The DM statistic for absolute

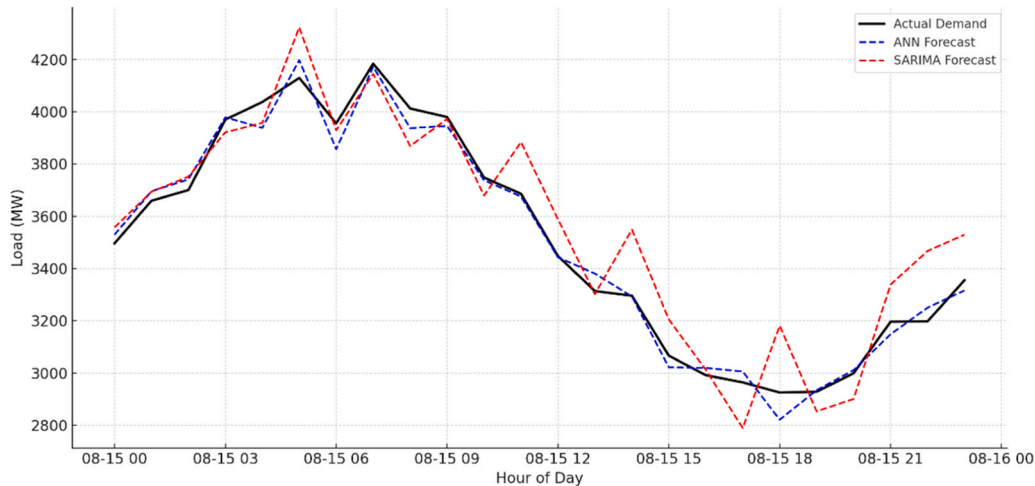


Fig. 4. ANN Architecture with Feature Importance and Statistical Performance. Source: Author.

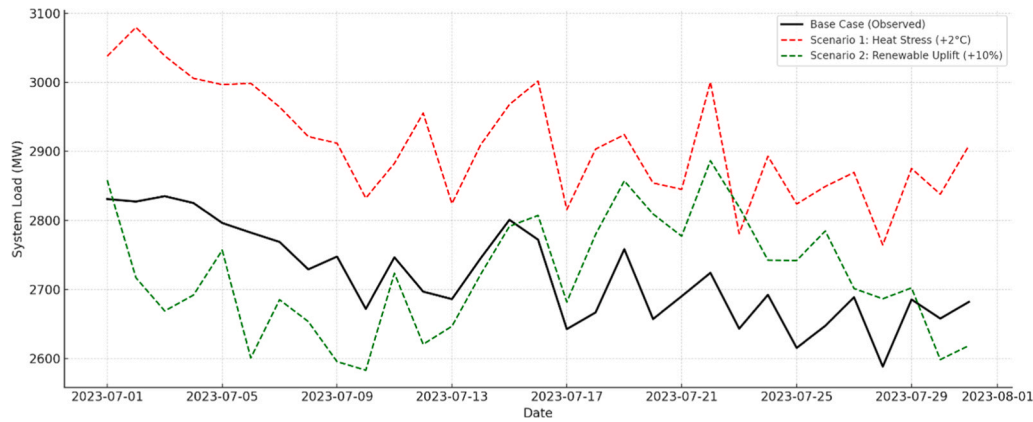


Fig. 5. Comparative Schematics: ANN vs. SARIMA with Statistical Results. Source: Author.

Table 3

Forecast Accuracy of ANN and SARIMA (Test Period: 2023–2024).

Horizon (Forecast Type)	Model	MAE (MW)	RMSE (MW)	MAPE (%)
Day-Ahead (Hourly)	SARIMA	162	228	4.67
	ANN	137	196	3.94
Week-Ahead (Daily)	SARIMA	146	201	4.09
	ANN	138	191	3.86
Month-Ahead (Monthly)	SARIMA	91	123	2.61
	ANN	97	132	2.76

Source: Author

errors of forecasts is 1.98 (p-value of 0.048). Over a month-ahead horizon, SARIMA slightly bettered the ANN, with a mean absolute percentage error (MAPE) of 2.61 % compared to 2.76 % for ANN, with a Diebold-Mariano (DM) test statistic of -2.12 ($p = 0.034$). These results support hypothesis 1 for short-term forecasting horizons, and SARIMA is found to be relevant for medium-term planning.

The second hypothesis states that the addition of exogenous variables, namely temperature, humidity, population, and gross domestic product (GDP) growth rate, would lead to improved accuracy in the forecasting models. For that purpose, comparisons are conducted for models with and without exogenous variables. In the context of the SARIMA framework, the addition of weather and demographic variables improved the mean absolute percentage error (MAPE) for a one-day ahead forecasting task by decreasing it from 5.31 % to 4.67 %. In the artificial neural network (ANN) framework, the addition of these variables while considering lag features of the electrical load improved the MAPE for the forecasting task by decreasing it from 4.28 % to 3.94 %. Paired *t*-tests conducted for the distributions of errors verified that these observations are statistically significant ($t = 3.27$, $p < .01$), validating the second hypothesis that these variables are crucial for improving the accuracy of forecasting models.

The third hypothesis states, “Increasing accuracy in demand forecasting will result in an improvement in the overall performance of the supply chain, including procurement, grid scheduling, and renewable integration.” This assumption can be supported by the results of the simulation, as it was observed that forecasts created through artificial neural networks (ANN) would save approximately 12 % on imbalance settlement costs as opposed to those created through SARIMA. This would amount to an actual savings of USD 4.2 million during the period from 2023 to 2024. It was also observed during grid scheduling simulations that, through ANN, approximately 11 % would be saved on the number of starts of peaking units, as well as approximately 7 % on ramping stress on combined-cycle plants. Reserve demand would also be reduced by approximately 130 MW, yet would remain equivalent.

The fourth hypothesis assumed that there was a positive relationship

between forecasting error and disruptions in the peak load. Analysis of the top 50 peak load hours for 2023–2024 indicated that, unlike SARIMA, which had a high level of forecasting error, an artificial neural network would have a decreased level of deviations in the peak demand. Furthermore, a logistic regression model that was developed to establish the relationship between the amount of error that occurred in the forecasts and the likelihood of disruptions indicated that there was a positive relationship, since the coefficient was 0.64, which was significant at 0.01. This indicated that when there was a high level of error, there was a high likelihood of disruptions. This was further supported when it was determined that, unlike SARIMA, which had issued nine peak demand warnings, the artificial neural network had issued five such warnings. This supported H4, which indicated that there was a relationship between improved forecasting accuracy and improved reliability.

The implications for the supply chain were examined using a deterministic operational simulation framework that converts errors in forecasts into imbalance volumes and reserves. In particular, errors in forecasts of demand in relation to actual demand were used for calculating additional balancing energies that are needed from peaking units or from reserves. The costs of imbalance were computed as being directly proportional to absolute errors in forecasts (in MW), time (in hours), and prevailing balancing costs. The reported 12 % reduction in imbalance costs is an expression of the improvement in absolute errors in forecasts obtained from ANN-based forecasts as opposed to SARIMA-based forecasts. Reserve requirements were computed using an approach that is standard for reliability-based calculations. In particular, increased uncertainties in forecasts demanded additional spinning reserves. Improved forecasts of demand reduced requirements for reserves, which in turn reduced dispatches of peaking units and operational costs. The benefits of renewable integration were computed from an analysis of net load deviations at increased solar penetrations. Improved forecasts of demand in the short term led to better alignment of dispatchable resources.

5. Discussion

The aspect of electricity demand forecasting forms a critical part in power system planning as well as in the study of energy economics. Time series-based approaches, such as the ARIMA as well as SARIMA models, have been effectively utilized in the process of forecasting the demand in power while considering the seasonal as well as trending effects in power consumption. Time series-based approaches have been observed to be effective in the process of forecasting, especially when stability in electrical consumption patterns has been evident [12,16]. For example, Ahmad and Jagatheswari [1] and Shehhi and Kaya [37] demonstrated the suitability of the SARIMA model for forecasting electrical consumption in the Gulf countries because of its capability for handling

annual seasonal patterns. However, other research indicates that these models become less effective in countries where climate change is affecting patterns [33].

For the aforementioned limitations, Artificial Neural Networks (ANNs) have been developed, particularly through their ability to handle nonlinear relations as they appear in electricity demand patterns [34]. There has been a great deal of research on the use of ANNs in the forecasting of energy demand, particularly in various contexts, including India and China, as presented by Wu et al. [50]. In the context of the Jordanian energy profile, where there are high levels of demand for cooling energy within the summer peak demand periods, there are similar non-linear patterns that can be observed. This similarity with other hot climate countries such as Saudi Arabia and Kuwait supports the use of non-linear approaches to energy demand forecasting [2,29].

Other than such methodological comparisons, there remains the question of what such improvements in forecasting accuracy achieve. In the field of supply chain management, for example, improved forecasting accuracy has been related to decreased bullwhip effects, improved efficiency in procurements, and improved agility in the allocation of resources [23,55]. Energy supply chains have similar characteristics, with the uncertainty feeding back into fuel procurements and forward into grid scheduling. In that respect, there is a similar integration with theory that was developed with respect to other types of supply chains, such as in the context of the importance of agility in responding to forecasts, as was discussed with respect to other sectors such as manufacturing or retailing [41]. There are additional complexities with respect to the operation of the electricity grid, which requires real-time operation and for which electricity is not storable.

According to the energy planning perspective, the Integrated Resource Planning (IRP) models have invariably indicated that the first input for generation expansion and investment decisions, with respect to generation capacity, has been the demand forecasts [11]. Any minor errors in the forecasting may lead to costly situations such as over- and under-capacity situations. Previous research has shown that, for transitioning economies such as Egypt and Morocco [40], the forecasting errors have impacted the renewable energy share and forced the use of costly peaking plants. Jordan's renewable energy expansion follows the forecasting accuracy and renewable share relationship. This relationship between forecasting accuracy and renewable share has also been observed in the European markets, wherein the increased variability in the net load has been observed with the increased share of renewable energy [39].

Moreover, yet another implication of this study pertains to the conceptualization of forecasting as a resource rather than a technical activity. The resource-based theory of strategic management views forecasting as a resource that improves organizational resilience [8]. Past research, such as that conducted on the energy sector, including that of Jaramillo et al. [24], has established that firms that utilize analytical forecasting systems not only minimize costs but also become strategically agile in response to policy, price, and or fuel price volatility. The resource-based theory of forecasting would be particularly applicable in Jordan, where it would be particularly useful to consider that Jordan relies heavily on imported fuels.

5.1. Managerial implications

The following are the implications of this study for policymakers, planners, and utility managers. Firstly, investments in infrastructure for better forecasting can result in cost savings by cutting down on imbalance charges, reserve requirements, and curtailing renewable energy sources. Secondly, for short-term planning, particularly for systems with high summer cooling demand, the benefits of the approach are greatest for neural network-based models of the interactions between weather and electricity demand. Thirdly, although ANN models are effective for short-term forecasting, the SARIMA models are also useful for medium-term planning and should not be ignored. In fact, the most effective

approach may be to use the strength of both models and develop a hybrid or ensemble approach. Fourthly, the forecasting capability should be viewed as a strategic organizational strength to cope with the uncertainties of fuel prices, climatic changes, and policy shifts. In the context of Jordan and other emerging economies, the inclusion of better demand forecasting techniques can improve the efficacy of the energy supply chain.

5.2. Supply chain implications

The improvement in the accuracy of the forecasts has provided significant benefits for the Jordan electricity supply chain. For instance, the improvement in the accuracy of the forecasts for the procurement stage, where the model switched from the SARIMA model to the ANN model, has seen the cost associated with the settlement of imbalances decline by about 12 %. This has meant a saving for the Jordan electricity supply chain of about USD 4.2 million for the two years under assessment. For the grid scheduling stage, the improvement in the accuracy of the forecasts has seen the number of peaking units decline by about 11 %, and the associated stress on the combined-cycle power stations decline by about 7 %. This has meant the reserve size requirement has been reduced by about 130 MW without any decline in the associated reliability standard, thereby increasing the active energy delivery potential. There has been significant improvement in the integration of renewables into the Jordan electricity supply chain, where the curtailment associated with the solar power generated has been reduced by about 5–7 %.

6. Conclusion

Within the context of forecasting in Jordan, the application of Artificial Neural Networks (ANNs) and the utilization of the Seasonal Autoregressive Integrated Moving Average (SARIMA) model have been identified as a key aspect in the strategic management of the energy SC. By drawing on the insights offered by the forecasting theory, the SCM literature, and the energy planning literature, the study establishes the contribution that the application of sophisticated forecasting tools can make to the efficiency, the integration of RESs, and the security of the emerging energy systems. With respect to the study's contribution to the field, it must be noted that the following issues were not explored within the scope of the current study: Firstly, the study did not have access to the granular smart meter demand and weather patterns that might have allowed the application of more accurate forecasting tools. Secondly, the study did not have the capacity to draw on the insights offered by probabilistic forecasting tools that might have allowed the application of a broader set of forecasting tools. Furthermore, the study did not have the capacity to draw on the insights offered by the application of forecasting tools that have the capacity to deal with the issue of breaks in the energy demand patterns that may be caused by policy reforms, the COVID-19 pandemic, and the integration of RESs. Lastly, the study did not have the capacity to draw on the insights offered by other forecasting tools that might have allowed the application of a broader set of forecasting tools.

CRedit authorship contribution statement

Badrea Al Oraini: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition. **Suleiman Ibrahim Mohammad:** Writing – review & editing, Visualization, Validation, Supervision. **Anber Abraheem Shlash Mohammad:** Resources, Project administration, Methodology, Funding acquisition. **Sultan Alaswad Alenazi:** Visualization, Supervision, Software, Methodology. **Torki M. Al-Fawwaz:** Visualization, Validation, Resources, Project administration. **Asokan Vasudevan:** Writing – review & editing, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

None

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References

- [1] Ahmad, S. Jagatheswari, Quantum safe multi-factor user authentication protocol for cloud-assisted medical IoT, *IEEE Access* 13 (2025) 3532–3545, <https://doi.org/10.1109/ACCESS.2024.3523530>.
- [2] A. Ahmad, S. Jagatheswari, Quantum-safe mutual authentication scheme for IoT using blockchain, *Results Eng.* 28 (2025) 106945, <https://doi.org/10.1016/j.rineng.2025.106945>.
- [3] A. Al-Abri, K.E. Okedu, Overview of the global electricity system in oman considering energy demand model forecast, *Energy Eng.* 120 (2) (2022) 409, <https://doi.org/10.32604/ee.2023.020375>.
- [4] F. Alasali, H. Foudeh, E.M. Ali, K. Nusair, W. Holderbaum, Forecasting and modelling the uncertainty of low voltage network demand and the effect of renewable energy sources, *Energies* 14 (8) (2021) 2151, <https://doi.org/10.3390/en14082151>.
- [5] M. Alawin, M. Al-Hamdi, M. Alomeri, Determinants of electricity demand in Jordan, *Indian J. Sci. Technol.* 9 (15) (2016), <https://doi.org/10.17485/ijst/2016/v9i15/88578>.
- [6] J. Allan, F. Mangili, M. Derboni, L. Gisler, A. Hainoun, A.E. Rizzoli, L. Ventriglia, M. Sulzer, A semantic data framework to support data-driven demand forecasting, *J. Phys. Conf. Ser.* 2600 (2) (2023) 22001, <https://doi.org/10.1088/1742-6596/2600/2/022001>.
- [7] H. Badr, W. Ahmed, A comprehensive analysis of demand prediction models in supply chain management, *Am. J. Ind. Bus. Manag.* 13 (12) (2023) 1353, <https://doi.org/10.4236/ajbm.2023.1312075>.
- [8] Y. Bogodistov, V. Wohlgenuth, Enterprise risk management: a capability-based perspective, *J. Risk Financ.* 18 (3) (2017) 234, <https://doi.org/10.1108/jrf-10-2016-0131>.
- [9] C. Boya, Analyzing the Relationship between Temperature and Load Demand in the Regions with the Highest Electricity Consumption in the Republic of Panama, 7th Int. Eng. Sci. Technol. Conf. (IESTEC) 2019 (2019) 132, <https://doi.org/10.1109/iestec46403.2019.00-88>.
- [10] J. Buitrago, S. Asfour, Short-Term Forecasting of Electric Loads Using Nonlinear Autoregressive Artificial Neural Networks with Exogenous Vector Inputs, *Energies* 10 (1) (2017) 40, <https://doi.org/10.3390/en10010040>.
- [11] J.P. Carvallo, P. Larsen, A.H. Stanstad, C. Goldman, Load. Forecast. Electr. Util. Integr. Resour. Plan. (2017), <https://doi.org/10.2172/1371722>.
- [12] H.-A.I. El-Azab, R.A. Swief, N.H. El-Amary, H.K. Temraz, Seasonal forecasting of the hourly electricity demand applying machine and deep learning algorithms impact analysis of different factors, *Sci. Rep.* 15 (1) (2025), <https://doi.org/10.1038/s41598-025-91878-0>.
- [13] M.M.A. Elgazzar, B. Tawfik, Time Series Analysis and Modeling of Monthly Rainfall in Saudi Arabia, *Int. J. Innov. Technol. Explor. Eng.* 9 (5) (2020) 1780, <https://doi.org/10.35940/ijitee.d1869.039520>.
- [14] A. Eshragh, B. Ganim, T. Perkins, K. Bandara, The Importance of Environmental Factors in Forecasting Australian Power Demand, *Environ. Model. Assess.* 27 (1) (2021) 1, <https://doi.org/10.1007/s10666-021-09806-1>.
- [15] S.N. Gowda, K. Nath, C. Zhang, R. Gowda, R. Gadh, Short-term aggregate electric vehicle charging load forecasting in diverse conditions with minimal data using transfer and meta-learning, *Energy Syst.* (2024), <https://doi.org/10.1007/s12667-024-00682-8>.
- [16] S. Haben, S. Arora, G. Giasemidis, M. Voß, D.V. Greetham, Review of low voltage load forecasting: Methods, applications, and recommendations, *Appl. Energy* 304 (2021) 117798, <https://doi.org/10.1016/j.apenergy.2021.117798>.
- [17] T.A. Hamed, L. Bressler, Energy security in Israel and Jordan: The role of renewable energy sources, *Renew. Energy* 135 (2018) 378, <https://doi.org/10.1016/j.renene.2018.12.036>.
- [18] D.D. Highley, T.J. Hilmes, Load forecasting by ANN, *IEEE Comput. Appl. Power* 6 (3) (1993) 10, <https://doi.org/10.1109/67.222735>.
- [19] Y. Hu, H. Liu, S. Wu, Y. Zhao, Z. Wang, X. Liu, Temporal collaborative attention for wind power forecasting, *Appl. Energy* 357 (2023) 122502, <https://doi.org/10.1016/j.apenergy.2023.122502>.
- [20] M.S. Hussain, K. Cho, S. Park, Resource Adequacy and Integration of Renewables in Light of US, EU, and Pakistan's Evolving Power Sector, *Energies* 17 (20) (2024) 5051, <https://doi.org/10.3390/en17205051>.
- [21] C.C. Ihebuchi, Day-Ahead Energy Price Forecasting with Machine Learning: Role of Endogenous Predictors, *Forecasting* 7 (2) (2025) 18, <https://doi.org/10.3390/forecast7020018>.
- [22] A.A. Ibrahim, B.N. Saeed, M.A. Fadel, Forecasting Stock Prices with an Integrated Approach Combining ARIMA and Machine Learning Techniques ARIMAML, *J. Comput. Commun.* 11 (8) (2023) 58, <https://doi.org/10.4236/jcc.2023.118005>.
- [23] Jahin, M.A., Shahriar, A., & Amin, M.A. (2024). MCDNF: Supply Chain Demand Forecasting via an Explainable Multi-Channel Data Fusion Network Model Integrating CNN, LSTM, and GRU. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2405.15598>.
- [24] M. Jaramillo, W. Pavón, L.F. Jaramillo, Adaptive Forecasting in Energy Consumption: A Bibliometric Analysis and Review, *Data* 9 (1) (2024) 13, <https://doi.org/10.3390/data9010013>.
- [25] R. Kalbasi, A. Tahmasebi, M.R. Ghaderi, M. Yari, F. Izadi, Toward sustainable energy-based buildings with focusing on electricity demand reduction – Case studies in Middle East region climate, *Sustain. Energy Technol. Assess.* 52 (2022) 102294, <https://doi.org/10.1016/j.seta.2022.102294>.
- [26] J. Liu, D. Zhang, S. Jiao, Z. Zhou, Z. Zhang, F. Gao, Preliminary study of radiative cooling in cooling season of the humid coastal area, *Sol. Energy Mater. Sol. Cells* 208 (2020) 110412, <https://doi.org/10.1016/j.solmat.2020.110412>.
- [27] R. Lotfi, A. Gholamrezaei, M. Kadlubek, M. Afshar, S.S. Ali, K. Kheiri, A robust and resilience machine learning for forecasting agri-food production, *Sci. Rep.* 12 (1) (2022), <https://doi.org/10.1038/s41598-022-26449-8>.
- [28] A. Louati, H. Louati, M. Alharbi, E. Kariri, A. Alkhonani, Forecasting trending jobs based on a Saudi's technological mega-corporation: STC Group, *Res. Sq. (Res. Sq.)* (2024), <https://doi.org/10.21203/rs.3.rs-4742501/v1>.
- [29] A. Malibari, A.H. Gamlo, Home Thermal Modeling: Cooling Energy Consumption and Costs in Saudi Arabia, *Comput. Inf. Sci.* 9 (4) (2016) 22, <https://doi.org/10.5539/cis.v9n4p22>.
- [30] A. Mystakidis, P. Koukaras, N. Tsalikidis, D. Ioannidis, C. Tjortjis, Energy Forecasting: A Comprehensive Review of Techniques and Technologies [Review of *Energy Forecasting: A Comprehensive Review of Techniques and Technologies*] (Multidisciplinary Digital Publishing Institute), *Energies* 17 (7) (2024) 1662, <https://doi.org/10.3390/en17071662>.
- [31] A. Nafil, M. Bouzi, K. Anoune, N. Ettalabi, Comparative study of forecasting methods for energy demand in Morocco, *Energy Rep.* 6 (2020) 523, <https://doi.org/10.1016/j.egyrs.2020.09.030>.
- [32] N. Neshat, H. Hadian, M. Behzad, Nonlinear ARIMAX model for long-term sectoral demand forecasting, *Manag. Sci. Lett.* (2018) 581, <https://doi.org/10.5267/j.msl.2018.4.032>.
- [33] S.A. Nugroho, A.F. Taha, How vintage linear systems controllers have become inadequate in renewables-heavy power systems: limitations and new solutions, *Am. Control Conf. (ACC) 2022* (2022) 4553, <https://doi.org/10.23919/acc53348.2022.9867388>.
- [34] D.-C. Park, M.A. El-Sharkawi, R.J. Marks, L. Atlas, M.J. Damborg, Electric load forecasting using an artificial neural network, *IEEE Trans. Power Syst.* 6 (2) (1991) 442, <https://doi.org/10.1109/59.76685>.
- [35] L. Schmid, M. Roidl, M. Pauly, Comparing statistical and machine learning methods for time series forecasting in data-driven logistics – A simulation study. *arXiv (Cornell University)*, 2023, <https://doi.org/10.48550/arxiv.2303.07139>.
- [36] A.A. Shahzad, A. Bamigbade, K. Gandhi, J.D. Barbosa, A. Sleptchenko, H. Apostoleris, S. Sgouridis, Realizing Renewable Resilience: Lessons from the Middle East for the Global Energy Transition, *iScience* 27 (7) (2024) 110058, <https://doi.org/10.1016/j.isci.2024.110058>.
- [37] M.R.A. Shehhi, A. Kaya, Time series and neural network to forecast water quality parameters using satellite data, *Cont. Shelf Res.* 231 (2021) 104612, <https://doi.org/10.1016/j.csr.2021.104612>.
- [38] A. Sinha, R. Tayal, A. Vyas, P. Pandey, O.P. Vyas, Forecasting Electricity Load With Hybrid Scalable Model Based on Stacked Non Linear Residual Approach, *Front. Energy Res.* 9 (2021), <https://doi.org/10.3389/fenrg.2021.720406>.
- [39] M.L. Sørensen, Forecasting and hierarchical reconciliation for renewable energy production and consumption, *Res. Portal Den.* 133 (2023). (<https://local.forskniisgportal.dk/local/dki-cgi/ws/cris-link?src=dtu&id=dtu-d871786b-03ee-4835-87e1-45ee9e7979c5&ti=Forecasting%20and%20hierarchical%20reconciliation%20for%20renewable%20energy%20production%20and%20consumption>).
- [40] H.G. Svendsen, A.A. Shetaya, K. Loudiyi, Integration of renewable energy and the benefit of storage from a grid and market perspective - results from Morocco and Egypt case studies, *Int. Renew. Sustain. Energy Conf. (IRSEC) 2016* (2016) 1164, <https://doi.org/10.1109/irsec.2016.7984007>.
- [41] A. Syntetos, Z. Babai, J.E. Boylan, S. Kolassa, K. Nikolopoulos, Supply chain forecasting: Theory, practice, their gap and the future, *Eur. J. Oper. Res.* 252 (1) (2015) 1, <https://doi.org/10.1016/j.ejor.2015.11.010>.
- [42] K. Szostek, D. Mazur, G. Dralus, J. Kuszniar, Analysis of the Effectiveness of ARIMA, SARIMA, and SVR Models in Time Series Forecasting: A Case Study of Wind Farm Energy Production, *Energies* 17 (19) (2024) 4803, <https://doi.org/10.3390/en17194803>.
- [43] C. Tarmanini, N. Sarma, C. Gezezin, O. Özgönenel, Short term load forecasting based on ARIMA and ANN approaches, *Energy Rep.* 9 (2023) 550, <https://doi.org/10.1016/j.egyrs.2023.01.060>.
- [44] J. Teng, H. Yin, Forecasting the carbon footprint of civil buildings under different floor area growth trends and varying energy supply methods, *Sci. Rep.* 13 (1) (2023), <https://doi.org/10.1038/s41598-023-49270-3>.
- [45] N. Tsalikidis, A. Mystakidis, C. Tjortjis, P. Koukaras, D. Ioannidis, Energy load forecasting: one-step ahead hybrid model utilizing ensemble, *Computing* 106 (1) (2023) 241, <https://doi.org/10.1007/s00607-023-01217-2>.
- [46] A. Ubaid, F.K. Hussain, M. Saqib, Container Shipment Demand Forecasting in the Australian Shipping Industry: A Case Study of Asia-Oceania Trade Lane, *J. Mar. Sci. Eng.* 9 (9) (2021) 968, <https://doi.org/10.3390/jmse9090968>.
- [47] L.C. Velasco, D. Lou, G. Paolo, M. Bryan, B. Felicísimo, A hybrid model of autoregressive integrated moving average and artificial neural network for load forecasting, *Int. J. Adv. Comput. Sci. Appl.* 10 (11) (2019), <https://doi.org/10.14569/ijacsa.2019.0101103>.

- [48] W. Waheed, Q. Xu, M. Aurangzeb, S. Iqbal, S.H. Dar, Z.M.S. Elbarbary, Empowering data-driven load forecasting by leveraging long short-term memory recurrent neural networks, *Heliyon* 10 (24) (2024), <https://doi.org/10.1016/j.heliyon.2024.e40934>.
- [49] X. Wen, J. Liao, Q. Niu, N. Shen, Y. Bao, Deep learning-driven hybrid model for short-term load forecasting and smart grid information management, *Sci. Rep.* 14 (1) (2024), <https://doi.org/10.1038/s41598-024-63262-x>.
- [50] C.-F. Wu, C.-M. Wang, T. Chang, C.-C. Yuan, The nexus of electricity and economic growth in major economies: The United States-India-China triangle, *Energy* 188 (2019) 116006, <https://doi.org/10.1016/j.energy.2019.116006>.
- [51] Y. Xiang, J. Liu, Y. Liu, Robust energy management of microgrid with uncertain renewable generation and load, *IEEE Trans. Smart Grid* (2015) 1, <https://doi.org/10.1109/tsg.2014.2385801>.
- [52] J. Yao, Electricity Consumption and Temperature, IMF Work. Pap. 21 (22) (2021), <https://doi.org/10.5089/9781513568539.001>.
- [53] H. Ye, X. Teng, B. Song, K. Zou, M. Zhu, Multi-source data fusion-based grid-level load forecasting, *Res. Sq. (Res. Sq.)* (2024), <https://doi.org/10.21203/rs.3.rs-5399298/v1>.
- [54] X. Zhang, F. Du, R. Hu, J. Li, F. Feng, Net load forecasting and energy storage demand analysis for renewable energy integration, *IET Conf. Proc.* 2023 (15) (2023) 1202, <https://doi.org/10.1049/icp.2023.2467>.
- [55] 宇都宮智昭, Optimization of cross-border E-commerce (CBEC) supply chain management based on fuzzy logic and auction theory, *Sci. Rep.* 14 (1) (2024), <https://doi.org/10.1038/s41598-024-64123-3>.